CS400M MINOR PROJECT

DevGenesis – AI-Powered Career and Code Intelligence Platform

SRS+SDS report

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NOV-2026

AI-Powered Web Application for Career Guidance, Codebase Intelligence, and Job Readiness

1. Introduction

Purpose

Students, fresh graduates, and early-career software developers currently rely on separate, disconnected platforms for resume evaluation, LinkedIn optimization, GitHub/portfolio analysis, job searching, interview preparation,
and skill development. This fragmentation makes it difficult for a candidate to understand their complete professional profile, identify weaknesses, and determine which companies and roles best suit them.

DevGenesis is being developed to solve this problem by providing a single, unified, AI-powered platform that analyzes a user's resume, LinkedIn profile, GitHub profile/repositories, projects, and skills together, and uses this combined understanding to deliver personalized career guidance, codebase intelligence, interview preparation, job recommendations, and skill-gap-driven learning roadmaps.

Scope

- The system will allow secure registration/login for users (email or OAuth-based).
- Users can upload resumes (PDF/DOCX) and link their LinkedIn and GitHub profiles for analysis.
- The system will generate resume/ATS scores, LinkedIn profile reviews, and GitHub/project health reports.
- Users can ask natural-language questions about their own codebase and receive context-aware answers using RAG.
- The system will conduct AI-based mock interviews (resume-based, project-based, DSA, system design, behavioral) and evaluate performance.
- The system will recommend suitable companies/roles based on the user's combined profile and track job applications.
- The system will identify skill gaps against target roles and generate personalized learning roadmaps.
- The system will generate a professional developer portfolio and structured, exportable project reports.
- The system will be accessible through any modern web browser (desktop and mobile responsive).

Users of the System

- **Registered User / Candidate —** uploads resume, connects LinkedIn/GitHub, takes AI interviews, receives recommendations and tracks applications.
- **Guest —** limited browsing of public features (e.g., landing page, sample reports) before registering.

Technology Stack

- **Frontend:** React.js with HTML, CSS, and JavaScript/TypeScript for an interactive dashboard and UI.
- **Backend:** Node.js with Express.js for core application APIs and business logic; Python with FastAPI for AI and document-processing microservices.
- **Database:** MongoDB, used to store user profiles, analysis results, interview records, job applications, and skill data.
- **AI/ML Layer:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), embeddings, and vector search for codebase QA, recommendations, and interview generation.
- **External Integrations:** GitHub API for repository access, and resume/document parsing for PDF/DOCX resumes.
- **Server/Hosting:** Node.js/Express application server with cloud hosting for the API, AI microservices, and database.

2. Software Requirements (SRS Part)

2.1 Functional Requirements

- **Authentication** — Secure registration and login (email/password and OAuth for GitHub/LinkedIn).
- **Resume & ATS Analysis** — Extract information from uploaded resumes and generate resume, ATS, and keyword-match scores with improvement suggestions.
- **LinkedIn Profile Review** — Analyze LinkedIn profile data and generate recommendations for improving professional presentation and recruiter visibility.
- **GitHub & Repository Analysis** — Analyze GitHub profile and repositories (structure, documentation, dependencies, APIs, database usage, authentication, testing, security) to generate project-health and recruiter-readiness insights.
- **Codebase Question Answering** — Allow users to ask questions about their own codebase and receive answers based on relevant repository context using RAG-based retrieval.
- **Project Explanation & Reports** — Generate structured project summaries, architecture explanations, and technical reports exportable as PDF.
- **AI Interview Simulator** — Conduct resume-based, project-based, codebase-based, DSA, CS-fundamentals, system-design, and behavioral mock interviews, and evaluate technical knowledge, problem-solving, communication, and behavioral responses.
- **Job Recommendation Engine** — Compare the user's combined professional profile with job requirements and recommend suitable companies/roles with direct application links where available.
- **Skill-Gap Analyzer** — Compare current skills against target-role requirements and generate a personalized learning roadmap.

- **Application Tracker** — Allow users to log and track the status of their job applications.
- **Portfolio Generator** — Generate a professional developer portfolio from the user's resume, projects, and GitHub data.
- **AI Career Twin** — Combine the user's professional information and performance data to provide a unified, continuously-updated view of career readiness with ongoing recommendations.

2.2 Non-Functional Requirements

- **Performance** — Standard pages should load in < 3 seconds; AI-generated responses (resume analysis, codebase QA, interview questions) should return within an acceptable time using asynchronous processing/loading indicators for longer AI tasks.
- **Security** — Passwords encrypted (hashing), secure OAuth token handling for GitHub/LinkedIn, role-based access control, and encrypted storage of sensitive resume/profile data.
- **Reliability** — Minimal downtime, consistent database access, and graceful handling of failures in external AI/API calls (retries, fallback messages).
- **Usability** — Easy navigation, responsive UI (desktop and mobile), and clear presentation of AI-generated scores/insights.
- **Scalability** — Should handle a growing number of users, repositories, and concurrent AI interview sessions through horizontally scalable backend/AI microservices.
- **Data Privacy** — User resumes, code, and profile data must only be accessed with explicit authorization and must not be shared with third parties without consent.

2.3 Constraints (as per requirement)

- Needs a stable internet connection for AI processing and GitHub/LinkedIn integration.
- Works best on modern browsers such as Chrome, Firefox, and Edge.
- Requires a hosting server with at least 4GB RAM and 2 CPU cores for the application backend; AI microservices may require additional compute/GPU resources depending on the LLM deployment.
- GitHub API and LLM usage are subject to third-party rate limits and usage costs.
- Analysis is limited to information the user explicitly authorizes access to (resume upload, GitHub/LinkedIn authorization).

3. System Design (SDS Part)

3.1 System Architecture

- **Client-Server Architecture with an AI microservice layer:**

[User/Browser] → [React.js Frontend] → [Node.js/Express API Layer] → [Python/FastAPI AI & RAG Services] ↔ [LLM / Vector Store] → [MongoDB Database]

The Node.js/Express layer handles authentication, business logic, and user-facing APIs, while the Python/FastAPI layer handles AI-driven tasks such as resume parsing, embeddings/RAG-based codebase QA, interview generation, and evaluation. Both layers read/write to a shared MongoDB database. External integrations (GitHub API, LinkedIn data, LLM providers) are accessed through the AI/backend services.

3.2 Module Design

- **Authentication Module** — Handles registration, login, and OAuth-based authorization for GitHub/LinkedIn.
- **Resume & ATS Module** — Parses uploaded resumes and generates scoring and improvement suggestions.
- **LinkedIn Analysis Module** — Reviews LinkedIn profile data and generates recommendations.
- **GitHub & Codebase Analysis Module** — Analyzes repositories and answers codebase questions using RAG/vector search.
- **Interview Simulator Module** — Generates and evaluates AI-driven mock interviews across multiple categories.
- **Job Recommendation Module** — Matches user profiles to job requirements and suggests roles/companies.
- **Skill-Gap & Roadmap Module** — Compares skills to target roles and generates learning roadmaps.
- **Application Tracker & Portfolio Module** — Tracks job applications and generates a shareable developer portfolio.
- **AI Career Twin Module** — Aggregates data from all modules to provide a continuously updated career-readiness view.
- **Database Module** — Stores and retrieves all user, analysis, interview, and job-related records.

3.3 Database Design

- **Entities:** User, Resume, LinkedInProfile, GitHubProfile, Repository, InterviewSession, JobApplication, SkillGap, LearningRoadmap, Portfolio.
- **ER Overview:** One User has one Resume, one LinkedInProfile, and one GitHubProfile (with many Repositories); a User can have many InterviewSessions and JobApplications; SkillGap/LearningRoadmap records are generated per User per target role.

Schema Example (Collections)

Collection	Key Fields
User	userId, name, email, passwordHash, role, targetRoles
Resume	resumeId, userId, fileUrl, atsScore, keywordScore, suggestions
GitHubProfile	userId, githubUsername, repositories[], projectHealthScore
InterviewSession	sessionId, userId, type, questions[], evaluationReport
JobApplication	applicationId, userId, company, role, status, appliedDate
SkillGap	userId, targetRole, missingSkills[], roadmap

3.4 User Interface Design

- **Wireframes / Screens:**
    - Login / Registration Page
    - User Dashboard (overview of resume, GitHub, LinkedIn, and career-readiness scores)
    - Resume & ATS Analyzer Page
    - GitHub/Codebase Analyzer & Chat (RAG Q&A) Page
    - AI Interview Simulator Page
    - Job Recommendations & Application Tracker Page
    - Skill-Gap & Learning Roadmap Page
    - Portfolio Generator Page
    - Admin Panel (user and data management)
- **Navigation Flow:** Login → Dashboard → Select Module (Resume / GitHub / Interview / Jobs / Skill-Gap / Portfolio) → View AI-Generated Insights → Logout.

3.5 Data Flow Diagrams (DFD)

- **Level 0:** Single process — "DevGenesis Platform" receiving inputs (resume, LinkedIn, GitHub data) from the User and returning career/codebase insights.
- **Level 1:** Expanded processes — Authentication, Resume/ATS Analysis, LinkedIn Analysis, GitHub/Codebase Analysis, Interview Simulation, Job Recommendation, Skill-Gap Analysis, all reading/writing to the central Database.

3.6 Sequence Diagram (Optional but good)

- **Example: Codebase Q&A Flow —**

User → Frontend → Backend API → AI/RAG Service (retrieves relevant repo context via vector search) → LLM (generates answer) → Backend → Frontend → User (Answer displayed).

4. Conclusion

- DevGenesis provides a secure, AI-powered, and scalable web-based platform that unifies resume, LinkedIn, GitHub, and job-readiness analysis into a single career-intelligence solution.
- Benefits: consolidated professional insights, automated codebase understanding, realistic AI-driven interview practice, targeted skill-gap guidance, and centralized job-application tracking.
- **Future Scope:** Mobile app integration, deeper AI-based company-fit recommendations, peer/mentor collaboration features, and multi-language support.

5. References

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